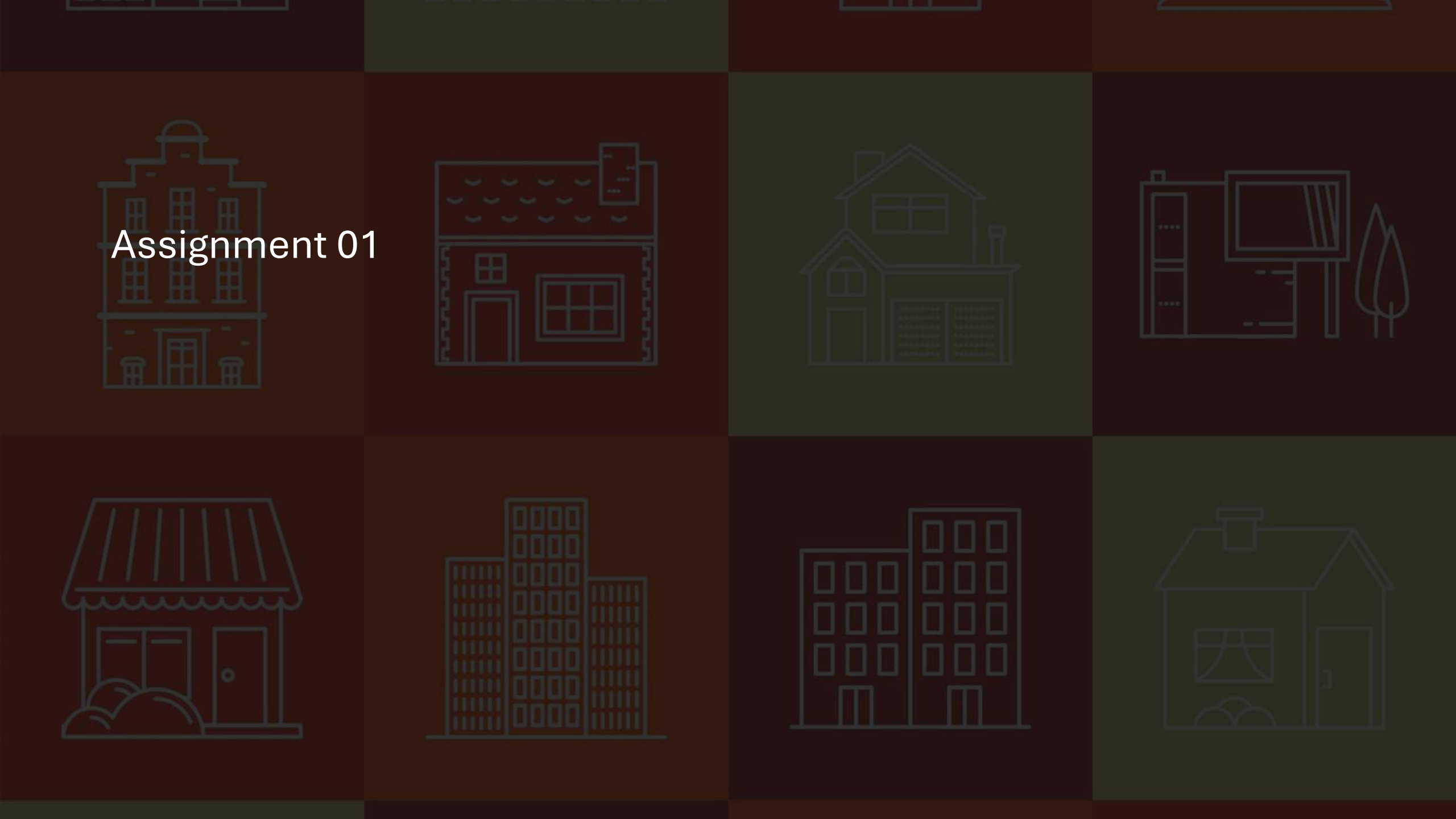
Assignment 01



Week 06
Regression Analysis

Highest level of education

Weekly working hours

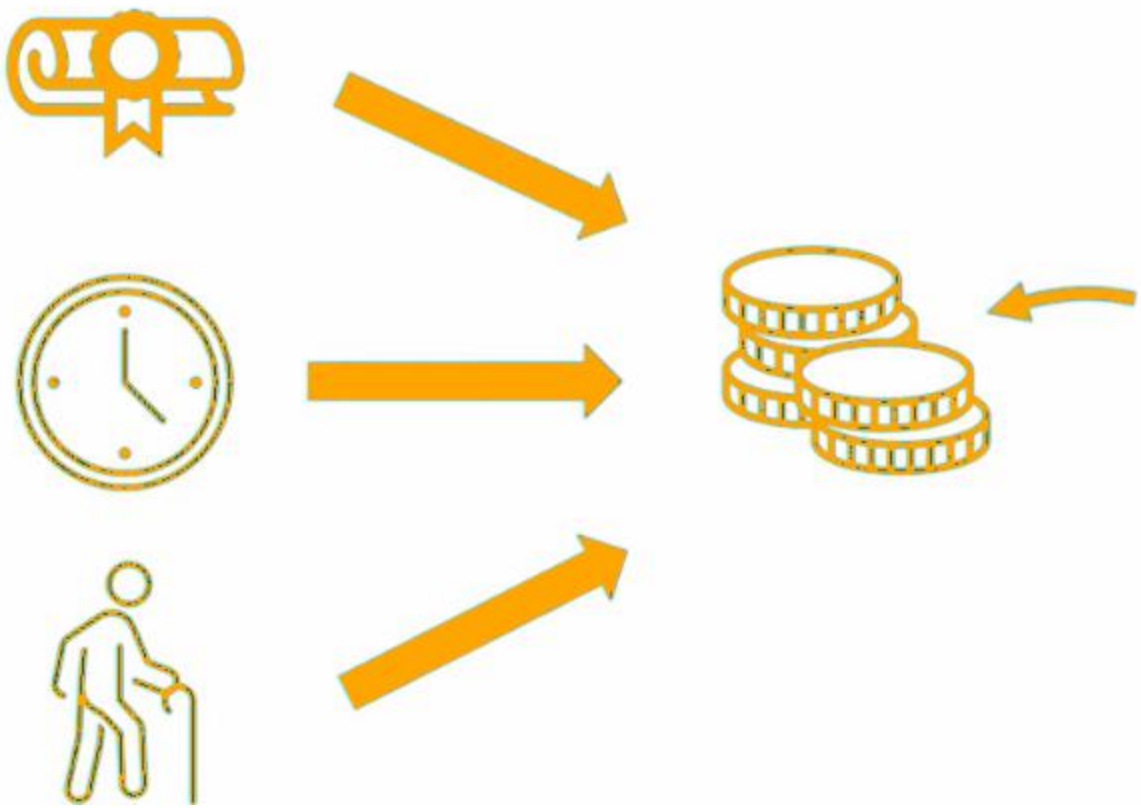
Age



What influences a person's **salary**?

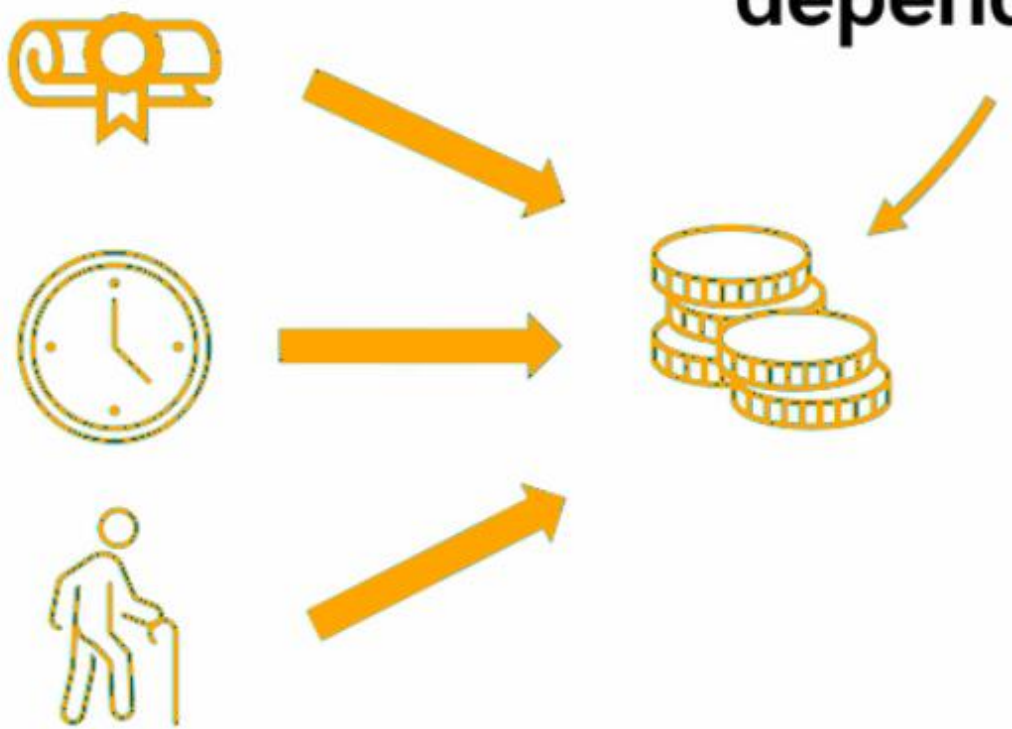


A **regression analysis** is a method for **modeling relationships** between **variables**.



It makes it possible to infer or predict a **variable**

The variable we want to infer or predict is called the **dependent variable**.



The variables we use for prediction are called **independent variables**.

Dependent versus independent variable



Regression analysis can be used to achieve two goals

Applications of regression analysis

Inferential analysis

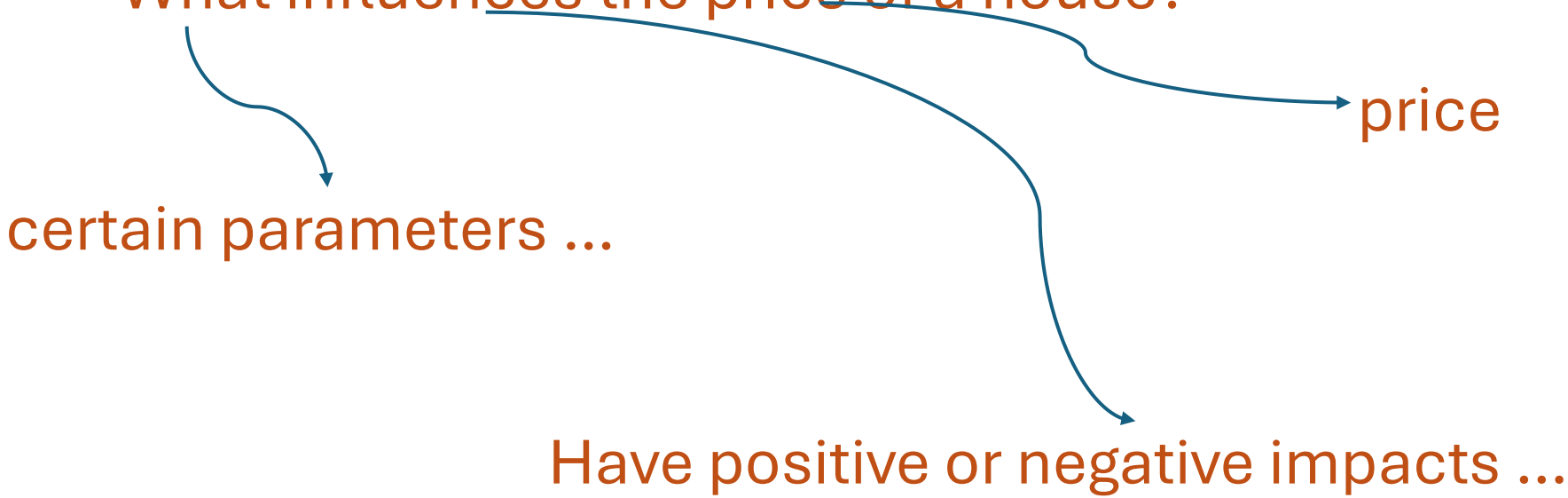
1- Measure the influence of one or more variables on another variable

Predictive analysis

2- Predict one variable based on one or more variables

1- Measure the influence of one or more variables on another variable

- What influences the price of a house?





BedroomA	KitchenAb	KitchenQu	TotRmsAb	Functional	Fireplaces	FireplaceQ	GarageTyp	GarageYrB	GarageFin	GarageCa	GarageAre	GarageQu	GarageCo	PavedDriv	WoodDeck	OpenPorch	EnclosedP	3SsnPorch	ScreenPor	PoolArea	PoolQC	Fence	MiscFeatu	MiscVal	MoSold	YrSold	SaleType	SaleCondit	SalePrice
3	1	Gd	8	Typ	0	NA	Attchd	2003	RFn	2	548	TA	TA	Y	0	61	0	0	0	0	NA	NA	NA	0	2	2008	WD	Normal	208500
3	1	TA	6	Typ	1	TA	Attchd	1976	RFn	2	460	TA	TA	Y	298	0	0	0	0	0	NA	NA	NA	0	5	2007	WD	Normal	181500
3	1	Gd	6	Typ	1	TA	Attchd	2001	RFn	2	608	TA	TA	Y	0	42	0	0	0	0	NA	NA	NA	0	9	2008	WD	Normal	223500
3	1	Gd	7	Typ	1	Gd	Detchd	1998	Unf	3	642	TA	TA	Y	0	35	272	0	0	0	NA	NA	NA	0	2	2006	WD	Abnorml	140000
4	1	Gd	9	Typ	1	TA	Attchd	2000	RFn	3	836	TA	TA	Y	192	84	0	0	0	0	NA	NA	NA	0	12	2008	WD	Normal	250000
1	1	TA	5	Typ	0	NA	Attchd	1993	Unf	2	480	TA	TA	Y	40	30	0	320	0	0	NA	MnPrv	Shed	700	10	2009	WD	Normal	143000
3	1	Gd	7	Typ	1	Gd	Attchd	2004	RFn	2	636	TA	TA	Y	255	57	0	0	0	0	NA	NA	NA	0	8	2007	WD	Normal	307000
3	1	TA	7	Typ	2	TA	Attchd	1973	RFn	2	484	TA	TA	Y	235	204	228	0	0	0	NA	NA	Shed	350	11	2009	WD	Normal	200000
2	2	TA	8	Min1	2	TA	Detchd	1931	Unf	2	468	Fa	TA	Y	90	0	205	0	0	0	NA	NA	NA	0	4	2008	WD	Abnorml	129900
2	2	TA	5	Typ	2	TA	Attchd	1939	RFn	1	205	Gd	TA	Y	0	4	0	0	0	0	NA	NA	NA	0	1	2008	WD	Normal	118000



In the previous example, we are not interested in predicting the price



Inferential analysis is very research-based and has many applications in social and economic science

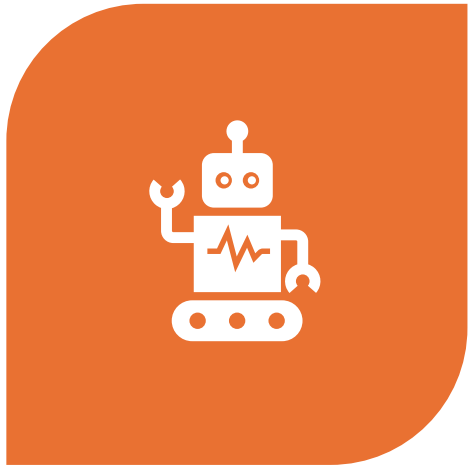
2- Predict one variable based on one or more variables

Predict the price of a house based on its property characteristics?



AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC
BedroomA	KitchenAb	KitchenQu	TotRmsAb	Functional	Fireplaces	FireplaceQ	GarageTyp	GarageYrB	GarageFin	GarageCa	GarageAre	GarageQu	GarageCo	PavedDrive	WoodDeck	OpenPorch	EnclosedP	3SsnPorch	ScreenPor	PoolArea	PoolQC	Fence	MiscFeatu	MiscVal	MoSold	YrSold	SaleType	SaleCondit	Price
2	1	TA	5	Typ	0	NA	Attchd	1961	Unf	1	730	TA	TA	Y	140	0	0	0	120	0	NA	MnPrv	NA	0	6	2010	WD	Normal	
3	1	Gd	6	Typ	0	NA	Attchd	1958	Unf	1	312	TA	TA	Y	393	36	0	0	0	0	NA	NA	Gar2	12500	6	2010	WD	Normal	
3	1	TA	6	Typ	1	TA	Attchd	1997	Fin	2	482	TA	TA	Y	212	34	0	0	0	0	NA	MnPrv	NA	0	3	2010	WD	Normal	
3	1	Gd	7	Typ	1	Gd	Attchd	1998	Fin	2	470	TA	TA	Y	360	36	0	0	0	0	NA	NA	NA	0	6	2010	WD	Normal	
2	1	Gd	5	Typ	0	NA	Attchd	1992	RFn	2	506	TA	TA	Y	0	82	0	0	144	0	NA	NA	NA	0	1	2010	WD	Normal	
3	1	TA	7	Typ	1	TA	Attchd	1993	Fin	2	440	TA	TA	Y	157	84	0	0	0	0	NA	NA	NA	0	4	2010	WD	Normal	
3	1	TA	6	Typ	0	NA	Attchd	1992	Fin	2	420	TA	TA	Y	483	21	0	0	0	0	NA	GdPrv	Shed	500	3	2010	WD	Normal	
3	1	TA	7	Typ	1	Gd	Attchd	1998	Fin	2	393	TA	TA	Y	0	75	0	0	0	0	NA	NA	NA	0	5	2010	WD	Normal	
2	1	Gd	5	Typ	1	Po	Attchd	1990	Unf	2	506	TA	TA	Y	192	0	0	0	0	0	NA	NA	NA	0	2	2010	WD	Normal	
2	1	TA	4	Typ	0	NA	Attchd	1970	Fin	2	525	TA	TA	Y	240	0	0	0	0	0	NA	MnPrv	NA	0	4	2010	WD	Normal	
2	1	Gd	5	Typ	1	Fa	Attchd	1999	Fin	2	511	TA	TA	Y	203	68	0	0	0	0	NA	NA	NA	0	6	2010	WD	Normal	
2	1	TA	5	Typ	0	NA	Detchd	1971	Unf	1	264	TA	TA	Y	275	0	0	0	0	0	NA	NA	NA	0	2	2010	COD	Normal	
3	1	TA	6	Typ	0	NA	Detchd	1997	Unf	1	320	TA	TA	Y	0	0	0	0	0	0	NA	NA	NA	0	3	2010	WD	Normal	
3	1	Gd	6	Typ	1	TA	Attchd	1975	Unf	2	440	TA	TA	Y	173	0	0	0	0	0	NA	NA	NA	0	6	2010	WD	Normal	

2- Predict one variable based on one or more variables



IS HIGHLY APPLICATION-ORIENTED



FOCUS IS TO PREDICT TO ENHANCE EFFICIENCY


plan

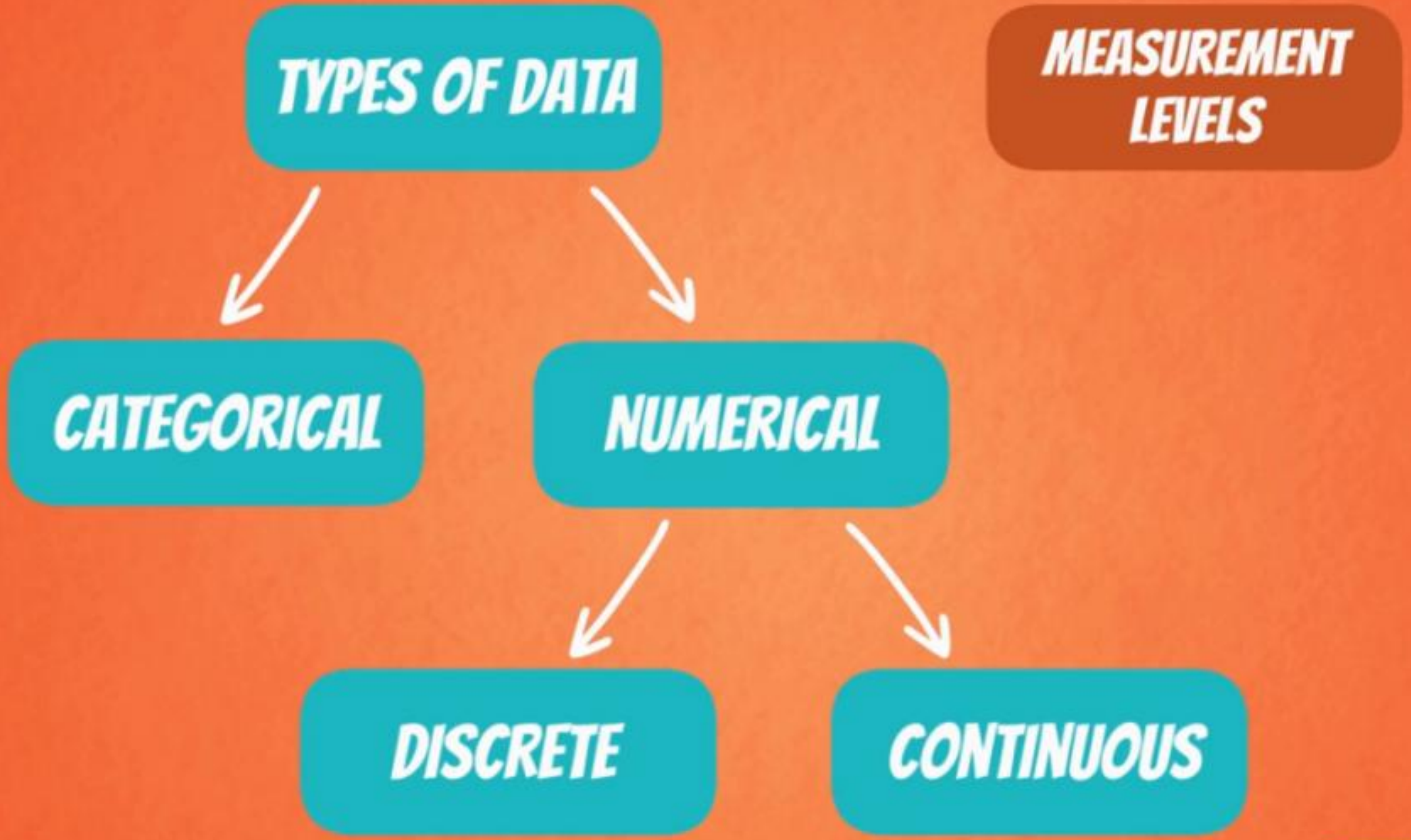
- In the next few weeks:

- First, we focus on regression analysis to measure the influence of one or more variables on another variable

- Next, we use regression analysis to predict one variable based on one or more variables

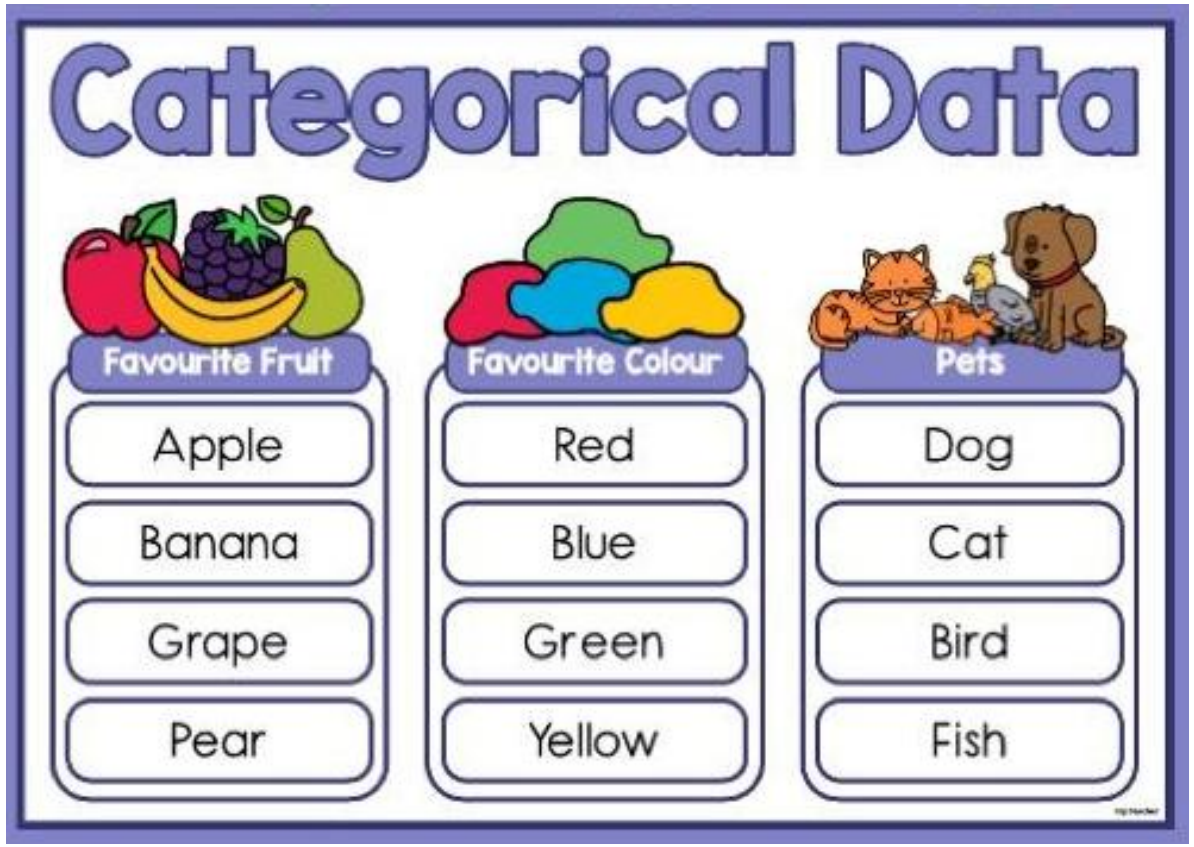
Different classes of regression analysis





365 DataScience

Categorical data



Categorical data

Logistic Regression



Risk of burnout



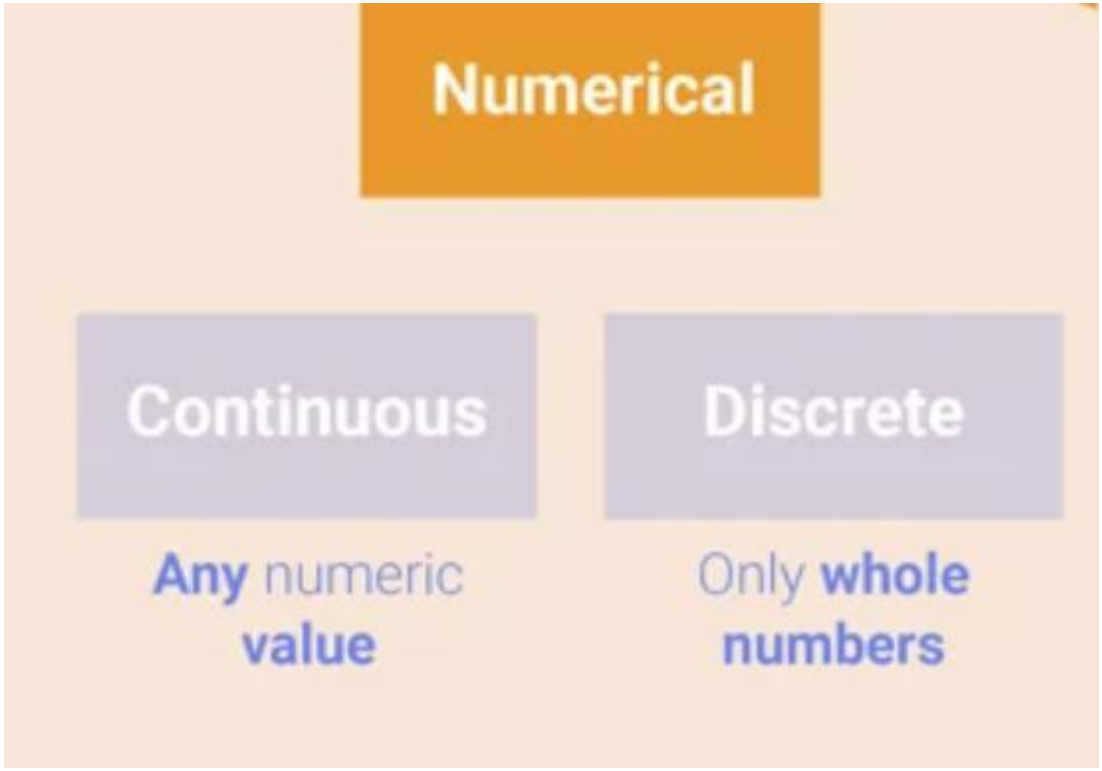
Diseased or not



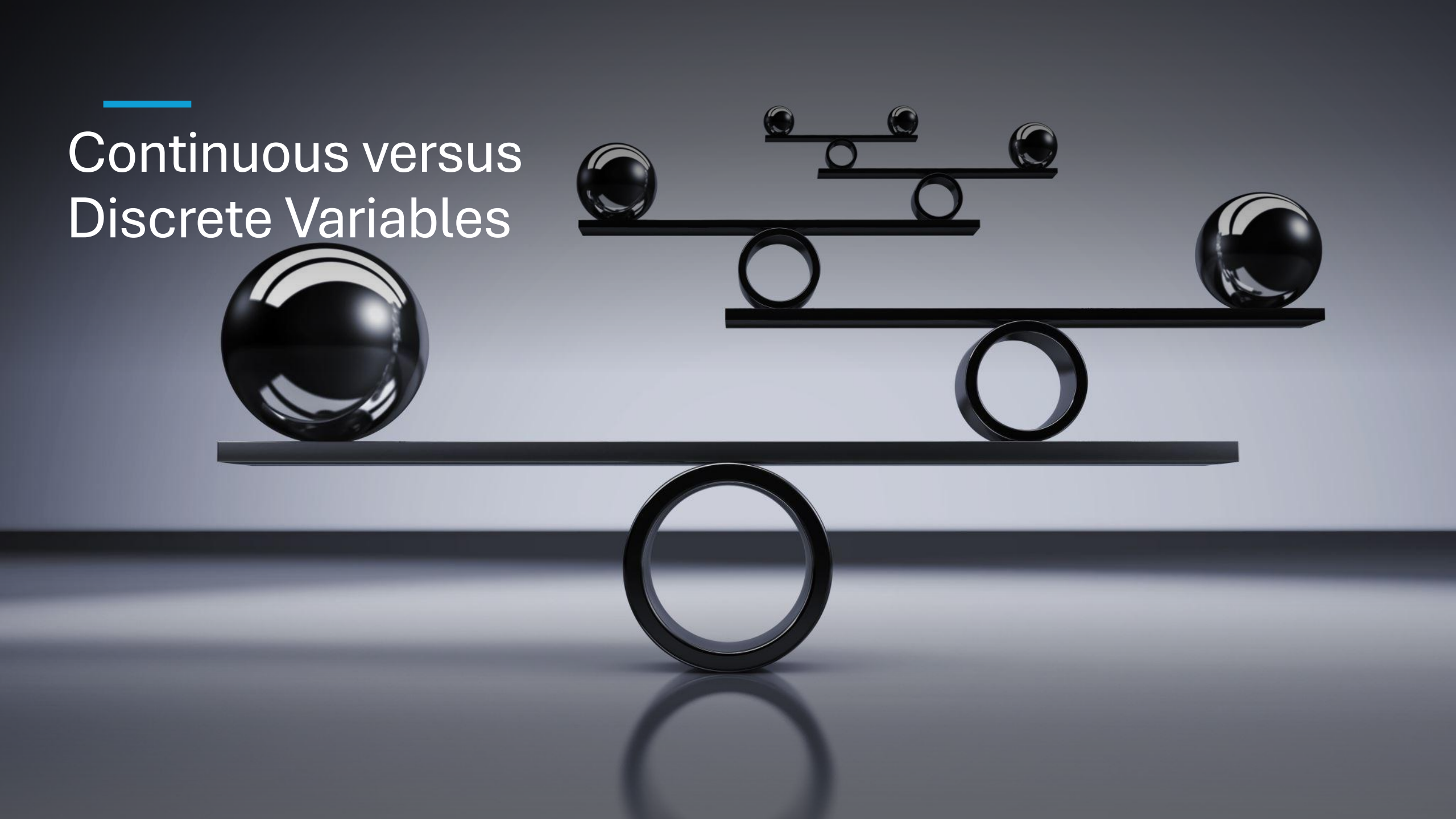
Animal



Numerical data



Continuous versus Discrete Variables



- Continuous variables represent measurable, infinite, and divisible values within a range (e.g., height, temperature).
- Discrete variables represent countable, distinct, and separate values (e.g., number of children).
- Discrete data cannot be broken into smaller portions, whereas continuous data can be made more exact

Discrete Data

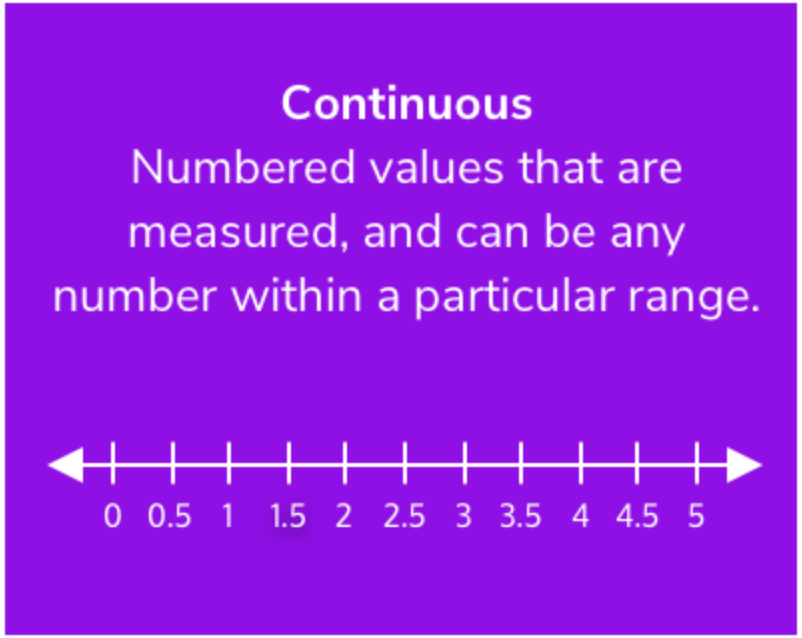
- 5 kids
- 96 workers
- 3 laptops

Continuous Data

- 3.25 kg
- 1.32 miles
- 7.25 inches

Linear regression

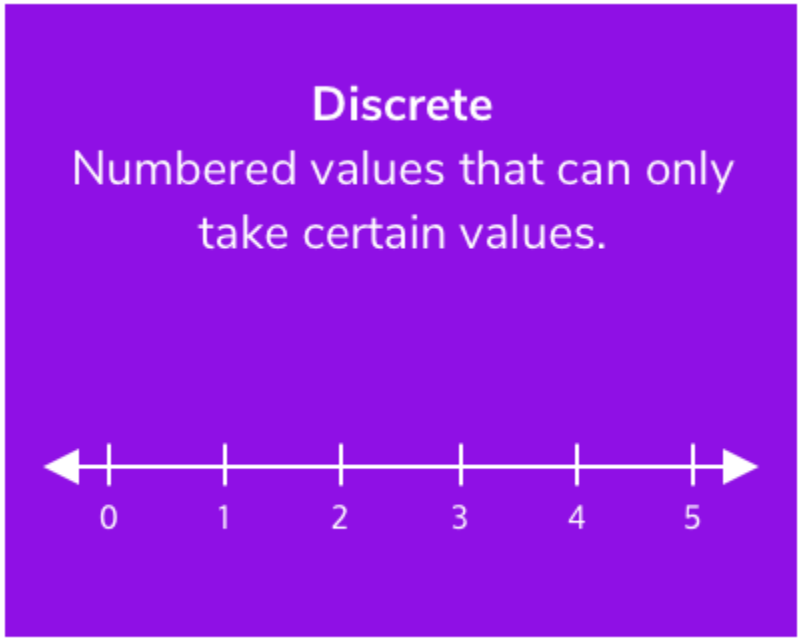
- **Condition**: continuous dependent variable



https://www.codecademy.com/learn/stats-variable-types/modules/stats-variable-types/cheatsheet

Poisson regression

Condition: discrete dependent variable



Linear regression

Correlation is not causation

- Causation describes a cause-and-effect relationship where one variable directly causes the other to change in a particular way.
- Correlation (Observation): When ice cream sales go up, the number of drowning incidents also goes up.
- Wrong Causation: "Eating ice cream causes drowning.”
- What is really happening (Actual cause): A third factor - hot weather - causes both: More people buy ice cream & More people go swimming → higher drowning



Linear Regression



Linear regression

The linear in linear regression indicates the relationship we can visualise as a line.

Linear regression is a technique that estimates the linear relationship between a **continuous dependent variable** y and **one or more independent variables** x.

E.g., what external factors affect the sales of our product? (e.g., size of the store, density of the neighborhood, demography of neighbourhood)

Linear regression



Two variables, x (IV) and y (DV), are related in a linear way, we say they are correlated.



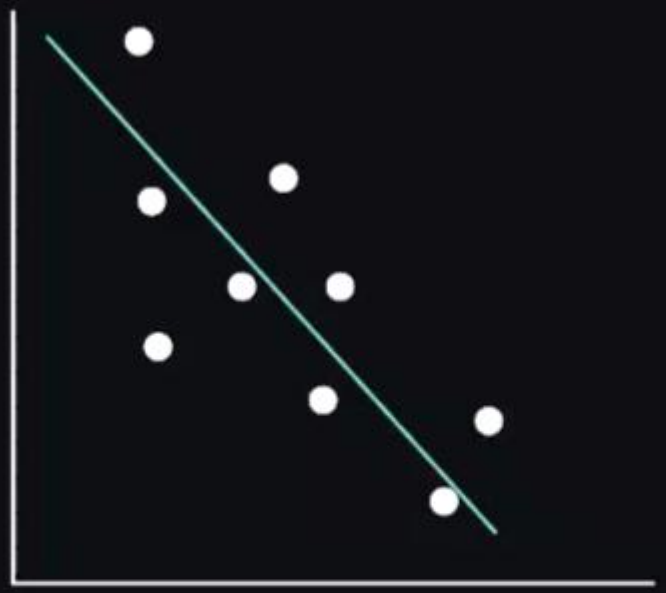
We calculate how strong the linear relationship between x and y is.

Linear regression – negative and positive correlation

Positive correlation: is a relationship between two variables that tend to increase or decrease together

Negative correlation: is an inverse relationship between two variables. When one variable increases, the other variable tends to decrease.

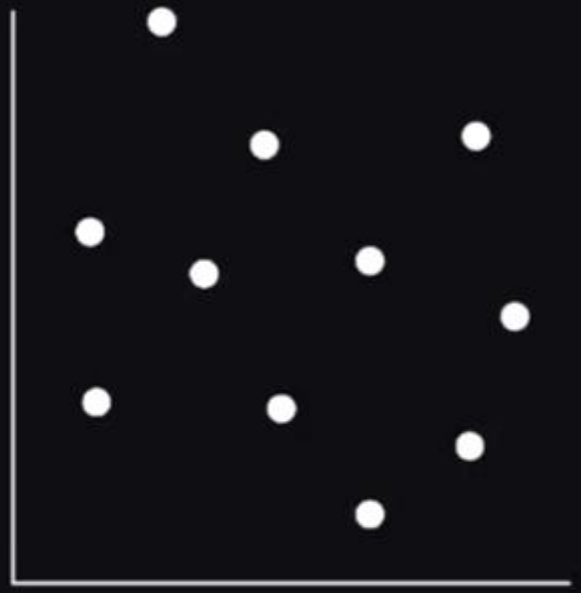
Negative Correlation



The plotted points are clustered around a downward sloping line.

As the values one variable decreases, the other tend to increase and vice versa.

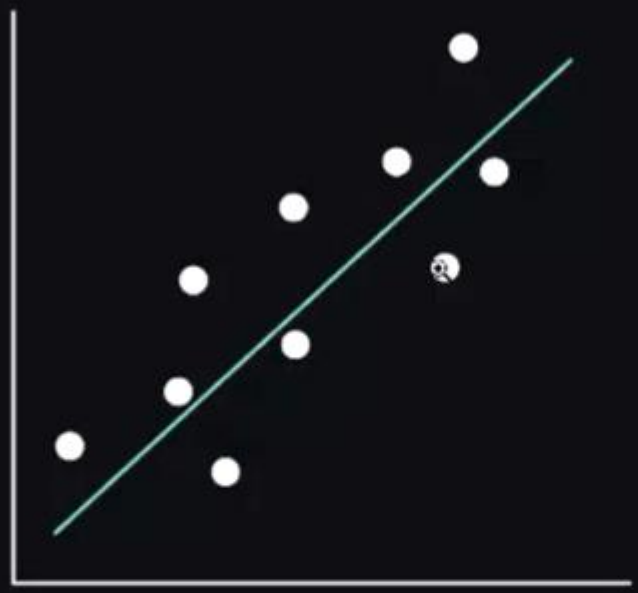
No Correlation



There is no linear trend.

There is no linear correlation between the two variables.

Positive Correlation

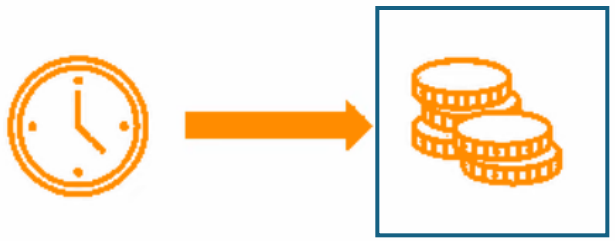


The plotted points are clustered around an upward sloping line.

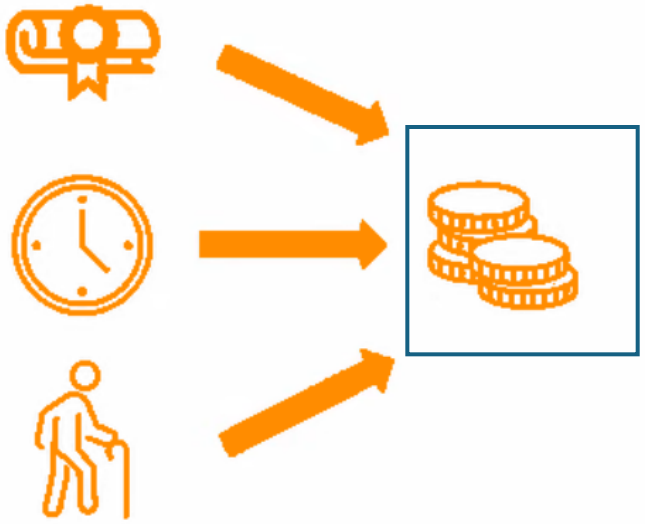
As the values of one variable increase, the value of the other variable tends to increase as well.
As the value of one variable decreases, the value of the other also decreases.

The dependent variable is **continuous**

Simple linear
Regression



Multiple linear
Regression



The dependent variable is **continuous**

Simple linear
Regression



Linear regression - sample

You want to understand the relationship between an author's social media following and their book sales. You would need data [ideally] on every book, social media follower count, and every book sale of all time.





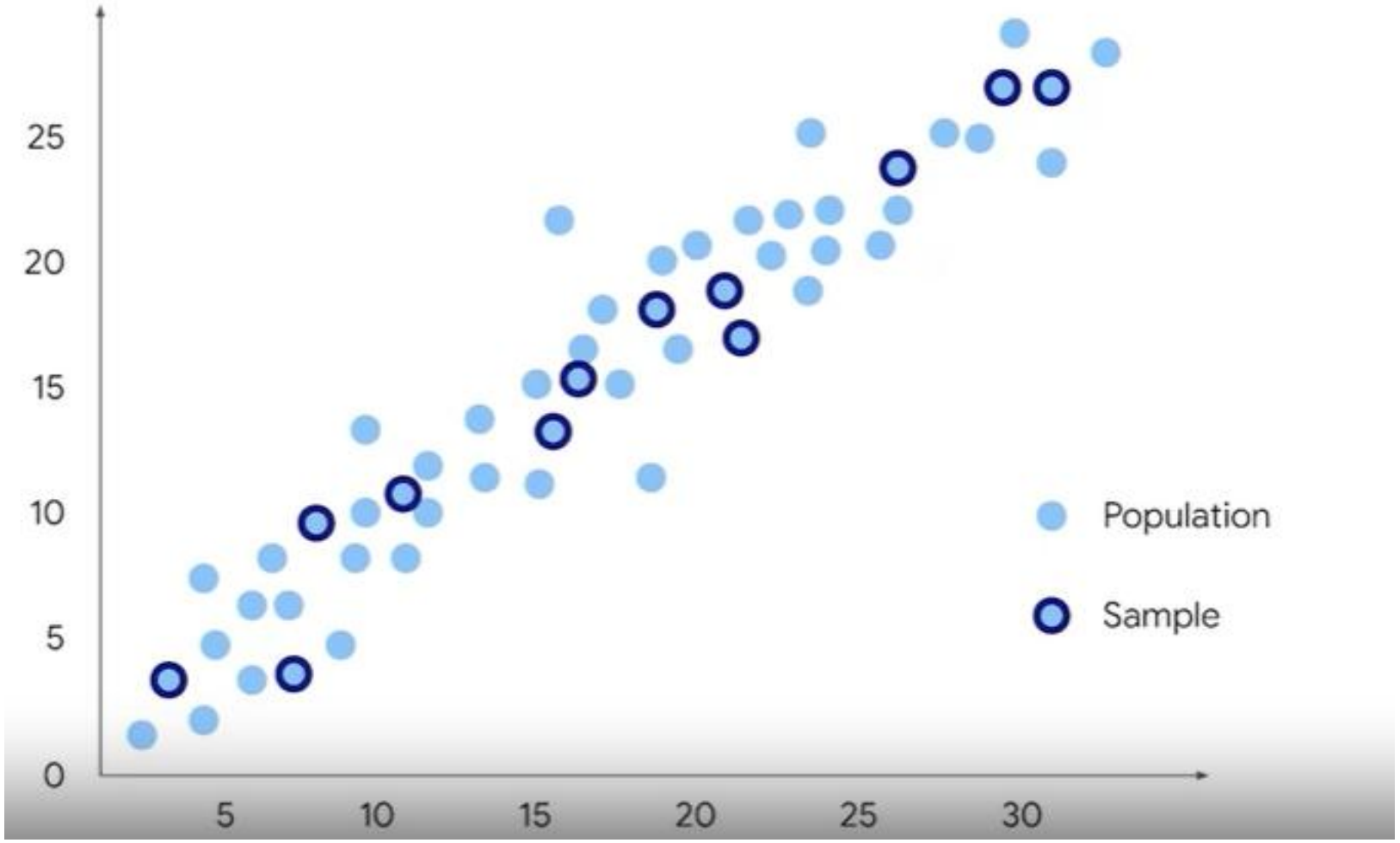
How to choose Y and X variables?

Linear regression - sample

You want to understand the relationship between an author's social media following and their book sales. You would need data [ideally] on every book, social media follower count, and every book sale of all time.



Linear regression - sample



Now we can use the collected data to calculate our regression model:

$$Y = slope * X + intercept$$

Sales amount

#social media followers

$$Y = slope * X + intercept$$

Linear regression – slope & intercept

The slope refers to the amount we expect y to increase or decrease per one unit increase of x, the independent variable.

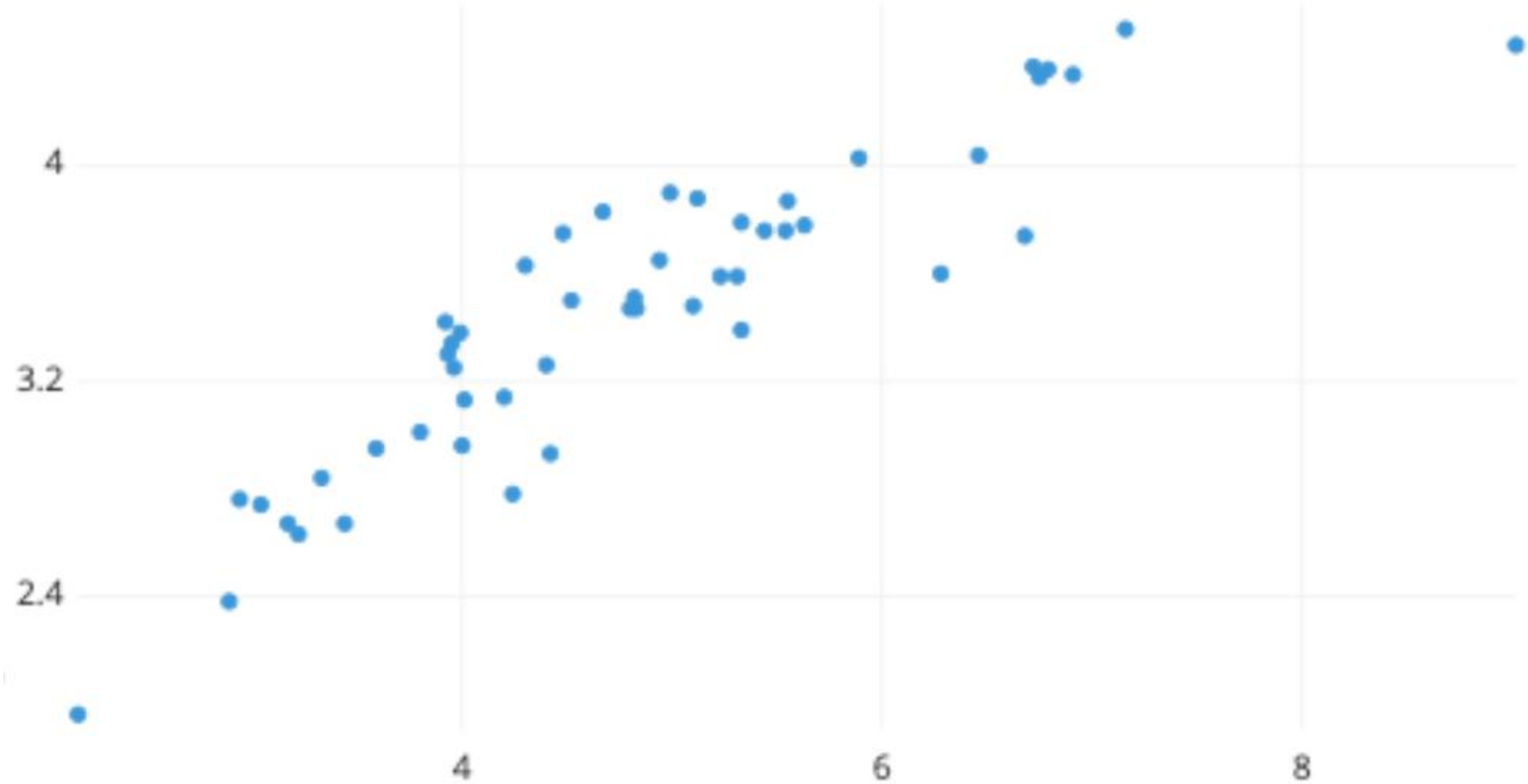
The intercept is the value of y when x equals 0.

Simple linear regression

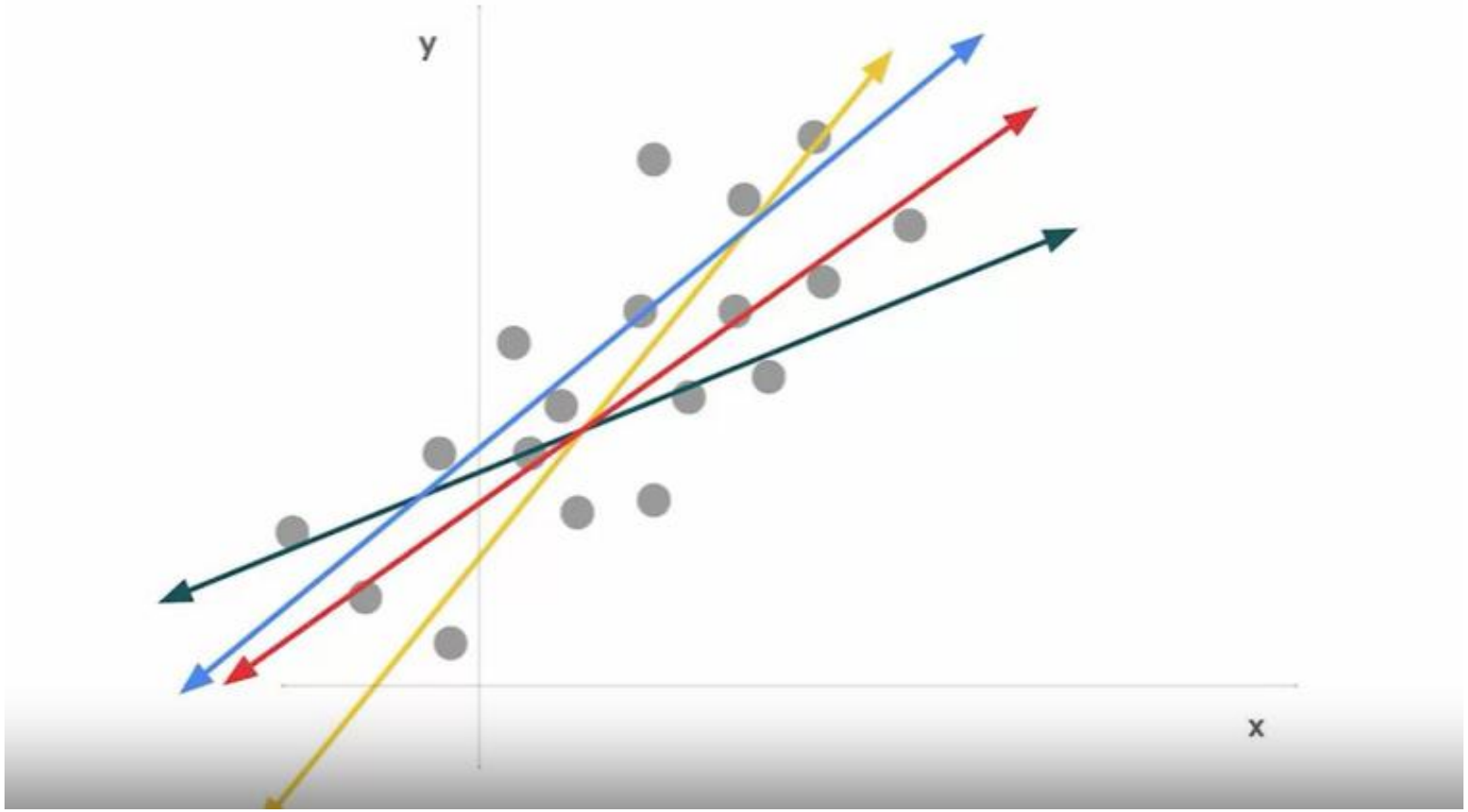
Linear regression indicates what the data looks like when plotted on an x, y. In simple linear regression, we're only interested in two variables, one x and one y variable.

$$y = \text{intercept} + \text{slope} * X$$

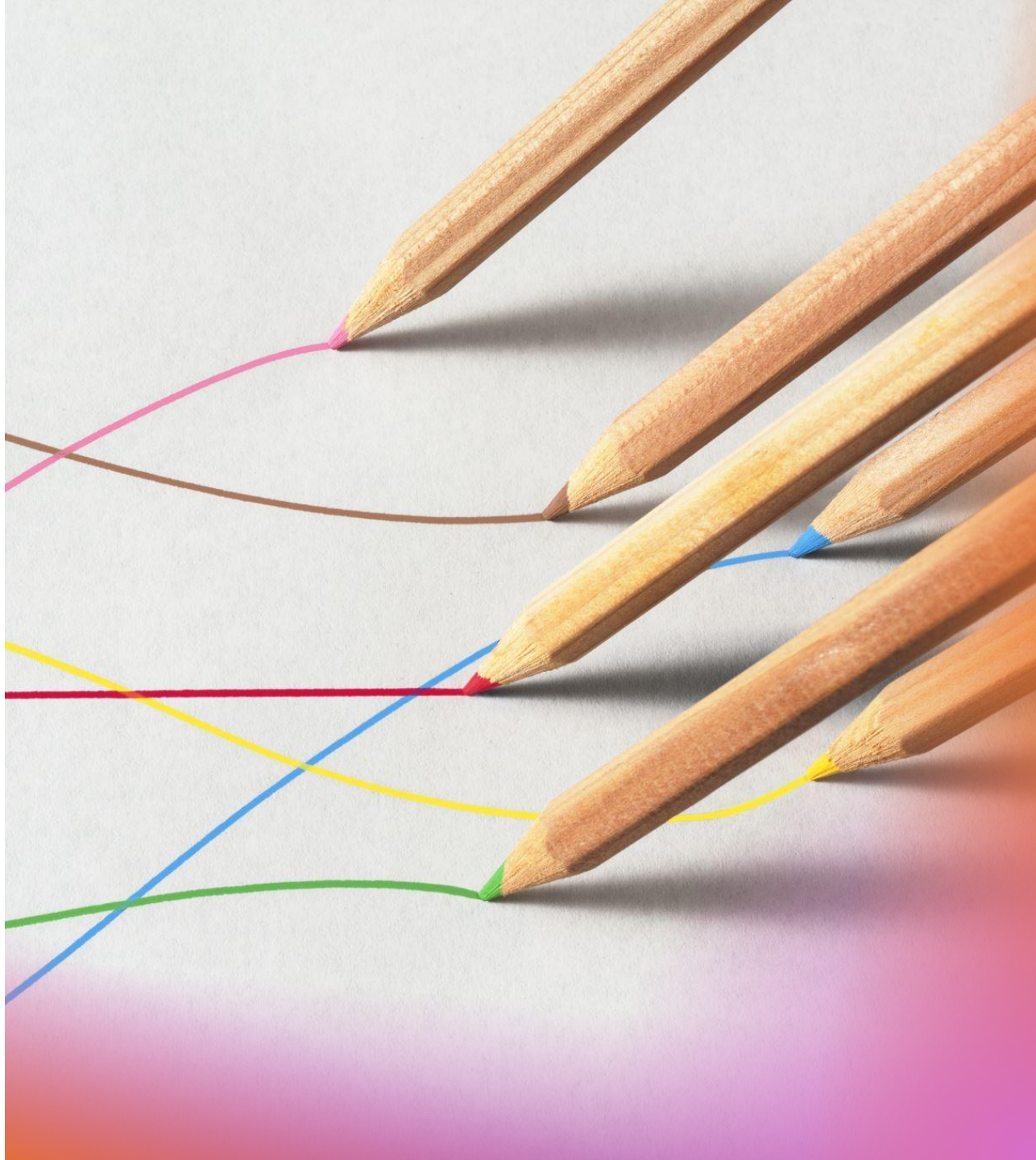
$$y = \beta_0 + \beta_1 X$$


4
3.2
2.4
4
6
8

Simple linear regression



How should we decide which regression line is the best formula for our regression model?



Ordinary Least Squares (OLS):

OLS is a method used in linear regression to estimate the coefficients (parameters) of the linear relationship between the dependent variable and one or more independent variables. **The primary goal of OLS is to find the line** that best fits the data by minimising the sum of the squared differences (residuals) between the observed values and the values predicted by the model (line).

In simpler terms, OLS finds the best-fitting line through a set of data points by ensuring that the differences between the actual data points and the predicted values (i.e., the errors or residuals) are as small as possible. The "least squares" part refers to the technique's focus on minimising the sum of the squared residuals.

Simple linear regression - residual

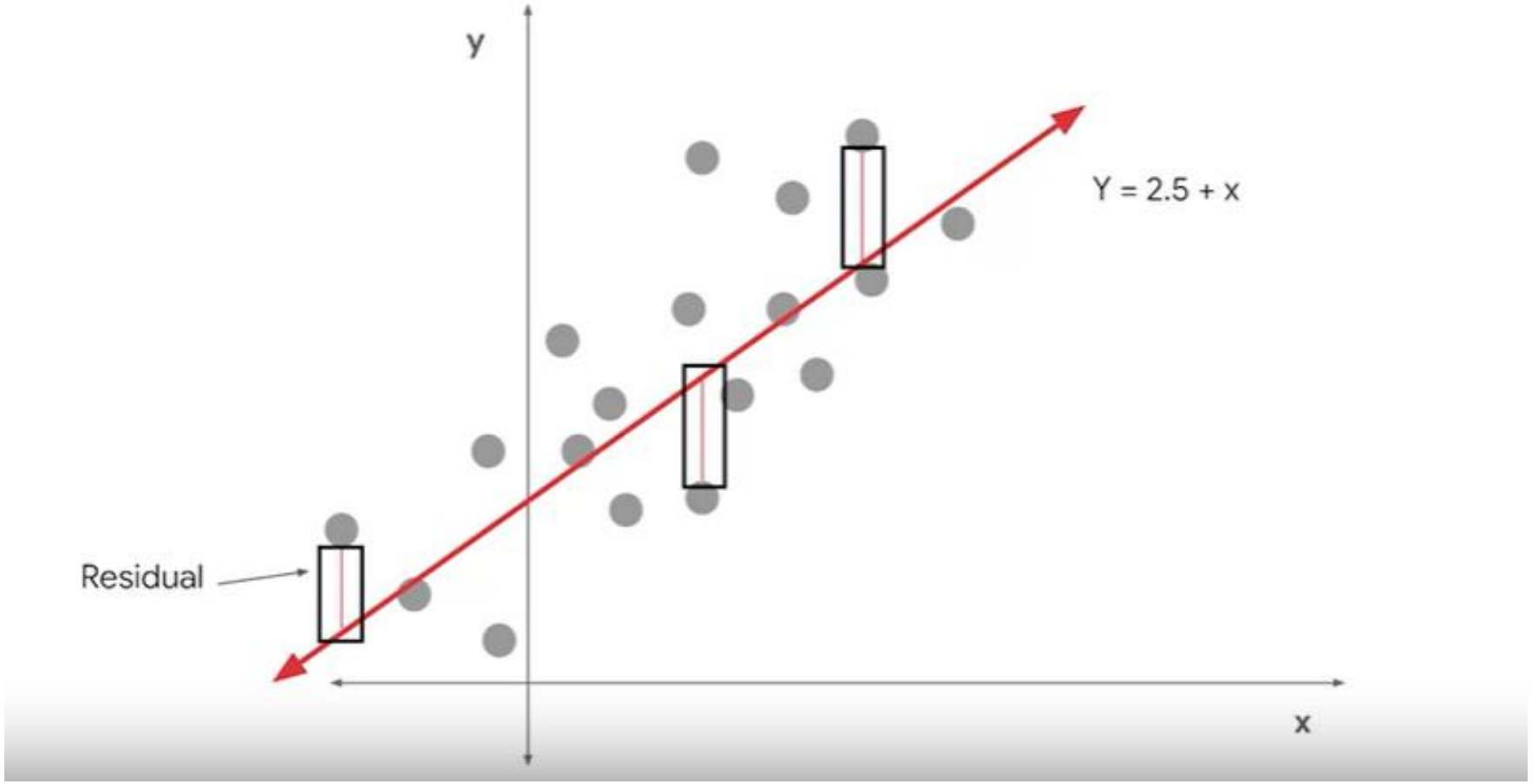
Residual = Observed - Predicted

Real data

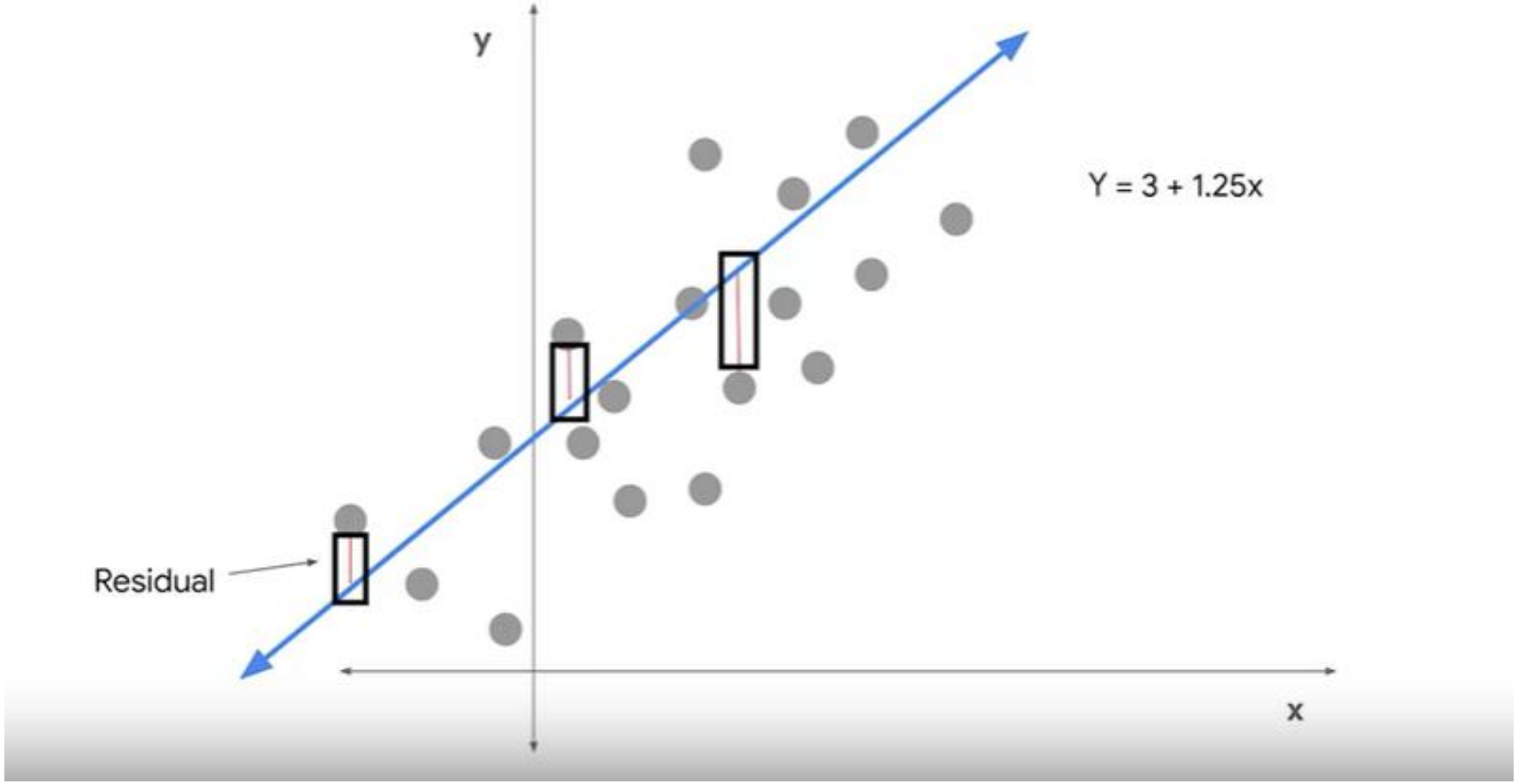
Data on the regression line

$$\varepsilon_i = y_i - \hat{y}_i$$

Simple linear regression – Ordinary Least Squares (OLS)

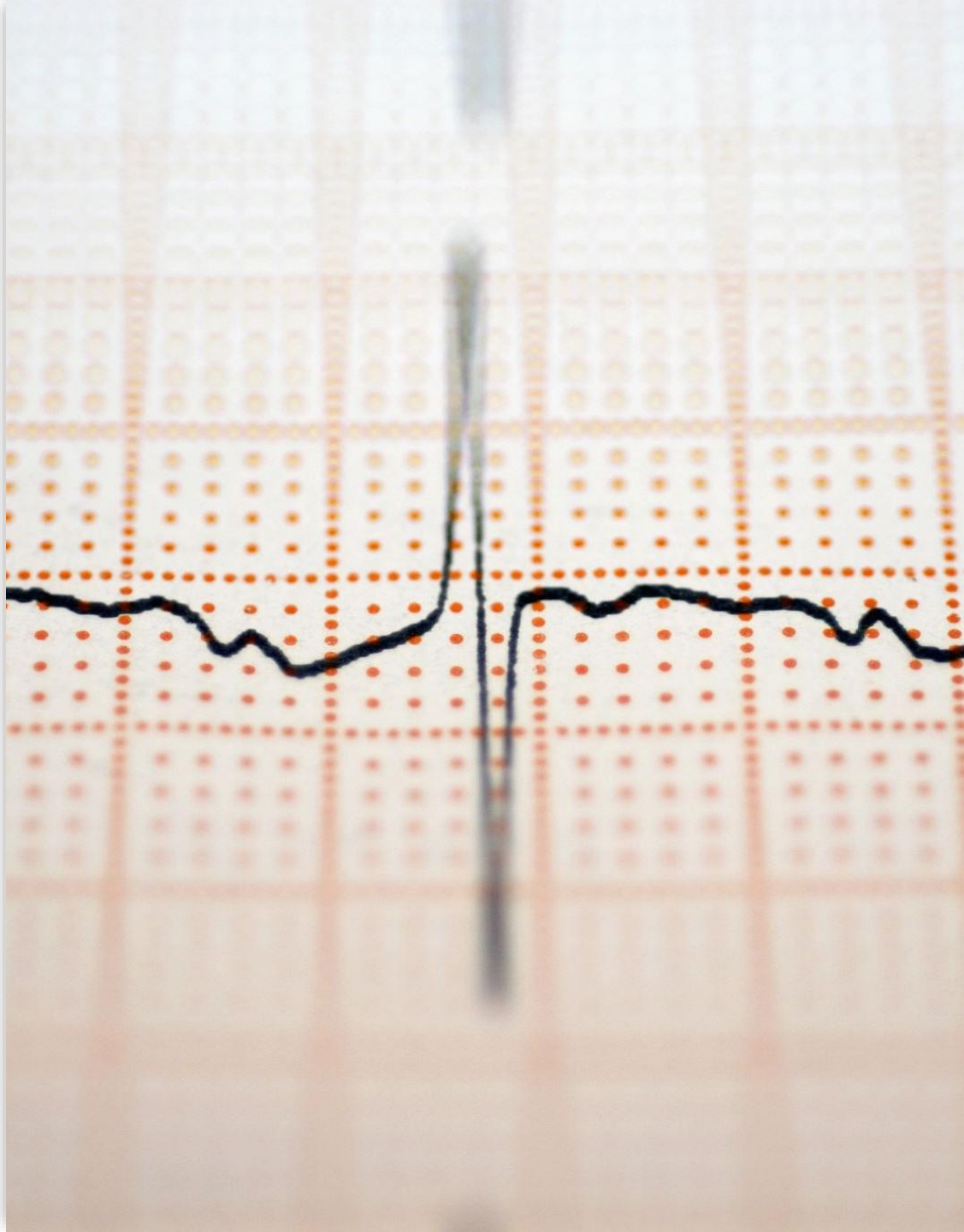


Simple linear regression – Ordinary Least Squares (OLS)



Confidence Intervals

A 95% confidence interval means that if we repeated the study many times, about 95% of the calculated values would contain the true value of the coefficient.



P-values

- When running regression analysis, you want to know if X is really correlated with y or not. So we do a hypothesis test on the regression results.

- With a 95 percent confidence rate, the p-value should be less than 0.05 to reject the null hypothesis. (i.e., no correlation).



R square

R-squared is a statistical measure in regression analysis that represents the proportion of the variance in the dependent variable that is explained by the independent variable(s) in the model. It is a key indicator of how well the regression model fits the observed data.

R^2 values range from 0 to 1.

$R^2 = 0$ indicates that the model explains none of the variance in the dependent variable.

In this case, the model does not fit the data at all.

$R^2 = 1$ indicates that the model explains all the variance in the dependent variable,

Simple linear regression – assumption summary

Data should be clean, e.g., no outliers, no skewness, no missing values

